

Day 10: RESTful APIs with Flask, Web Scraping & Introduction to NumPy

Welcome to Day 10! Today's session covers three practical and foundational topics in modern Python development:

1. **RESTful Web Services using Flask:** Building decoupled, stateless JSON APIs that power Single Page Applications (React, Angular, Vue), mobile applications, and distributed microservices.
 2. **Web Scraping Basics in Python:** Extracting structured data from web pages when no official API exists, using `requests` and `BeautifulSoup`.
 3. **Introduction to NumPy (Image Manipulation):** Understanding high-performance numerical computing with `ndarray` and manipulating digital images directly as multi-dimensional arrays.
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SECTION A: Fundamentals of REST API using Flask

Part 1: Architectural Foundations of REST

1. What is an API?

An **Application Programming Interface (API)** is a formal contract between two software systems defining how they communicate, what requests can be made, what data formats must be supplied, and what response formats will be returned.

In modern computing, APIs form the connective tissue between:

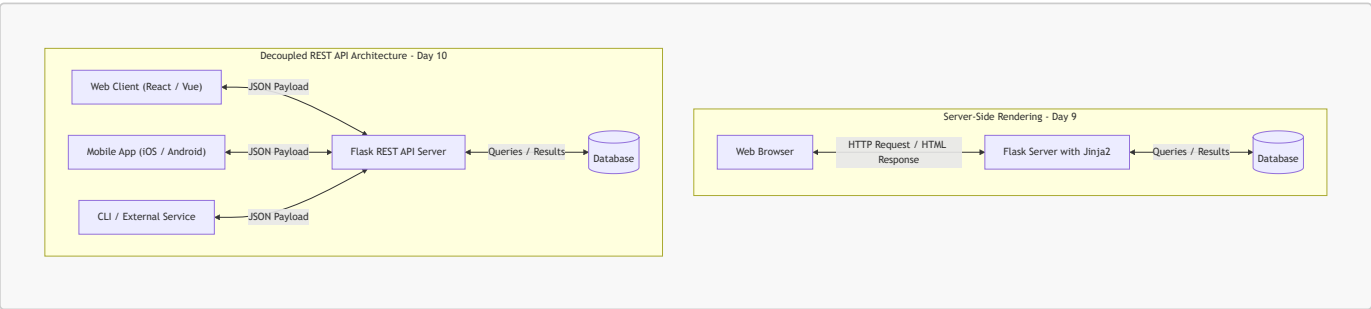
- Frontend user interfaces (React, Vue, iOS, Android) and backend servers.
- Different microservices running inside a cloud ecosystem (e.g., Payment Service talking to Order Service).
- Third-party integrations (e.g., using Stripe API for payments, Twilio for SMS, Google Maps API for geolocation).

2. Server-Side Rendering (SSR) vs. REST API Decoupling

In Day 9, we built **Server-Side Rendered (SSR)** applications where Flask generated complete HTML pages using Jinja2 templates. When a client asked for data, the server queried the database, merged data into HTML, and transmitted raw HTML markup back to the browser.

In contrast, **REST APIs decouple presentation from data**:

- The server sends **raw data** (typically JSON payloads).
- The client (browser, mobile app, desktop app, or IoT device) receives this data and is solely responsible for how it gets rendered.



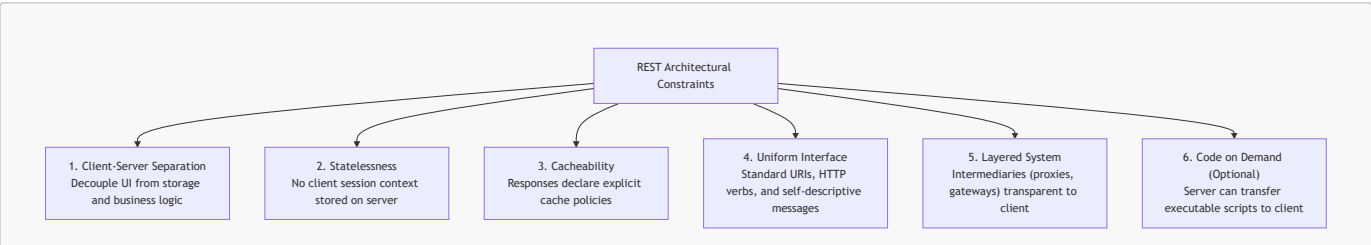
Advantages of the REST API Architecture:

1. **Multi-Client Support:** A single Flask backend API can simultaneously serve a web app, an iOS app, an Android app, a CLI tool, and partner integrations.
2. **Bandwidth Efficiency:** Instead of sending 50 KB of repetitive HTML markup, the server sends a 2 KB JSON packet containing only raw values.
3. **Separation of Concerns:** Backend engineers focus exclusively on database efficiency, security, transactions, and business logic. Frontend engineers focus on user experience, styling, accessibility, and UI performance.

3. Roy Fielding's 6 REST Architectural Constraints

The term **REST** stands for **Representational State Transfer**. It was introduced in 2000 by computer scientist **Roy Fielding** in his doctoral dissertation "*Architectural Styles and the Design of Network-based Software Architectures*".

To be truly **RESTful**, a system must adhere to six architectural constraints:



1. Client-Server Separation:

- The user interface concerns are separated from the data storage and business logic concerns.
- This improves user interface portability across multiple platforms and allows backend components to scale independently.

2. Statelessness:

- **Crucial Rule:** The server must not store any session context about the client between requests.
- Every incoming request must contain **all** the information necessary for the server to authenticate, authorize, and fulfill it (e.g., via API keys, JWT bearer tokens, or authorization headers).
- **Benefit:** Extreme scalability. Any incoming request can be handled by any server instance in a load-balanced cluster without session synchronization.

3. Cacheability:

- Responses must define themselves as cacheable or non-cacheable using standard HTTP headers (**Cache-Control**, **ETag**, **Expires**).
- If a response is cacheable, intermediate proxies or client browsers are permitted to reuse that response data for equivalent subsequent requests, reducing latency and network traffic.

4. Uniform Interface:

- The central constraint that distinguishes REST from other network architectures. It comprises four sub-principles:
 - **Identification of Resources:** Every conceptual entity (e.g., a product, a customer) is identified with a unique URI (e.g., `/api/v1/products/42`).
 - **Manipulation of Resources through Representations:** When a client holds a representation of a resource (e.g., JSON), it has enough information to modify or delete the resource on the server (given adequate permissions).
 - **Self-Descriptive Messages:** Each message includes enough metadata (like **Content-Type: application/json**) describing how to process the body.
 - **Hypermedia As The Engine Of Application State (HATEOAS):** Clients make state transitions dynamically by traversing hypermedia links provided within the response payloads.

5. Layered System:

- The architecture is composed of hierarchical layers (e.g., reverse proxies, load balancers, API gateways, firewall layers).
- A client cannot tell whether it is communicating directly with the end server or with an intermediary along the path.

6. Code on Demand (Optional):

- Servers may temporarily extend or customize client functionality by transferring executable code (e.g., JavaScript scripts or compiled WebAssembly applets).

Part 2: RESTful URI Design & HTTP Semantics

1. Resource-Oriented URI Conventions

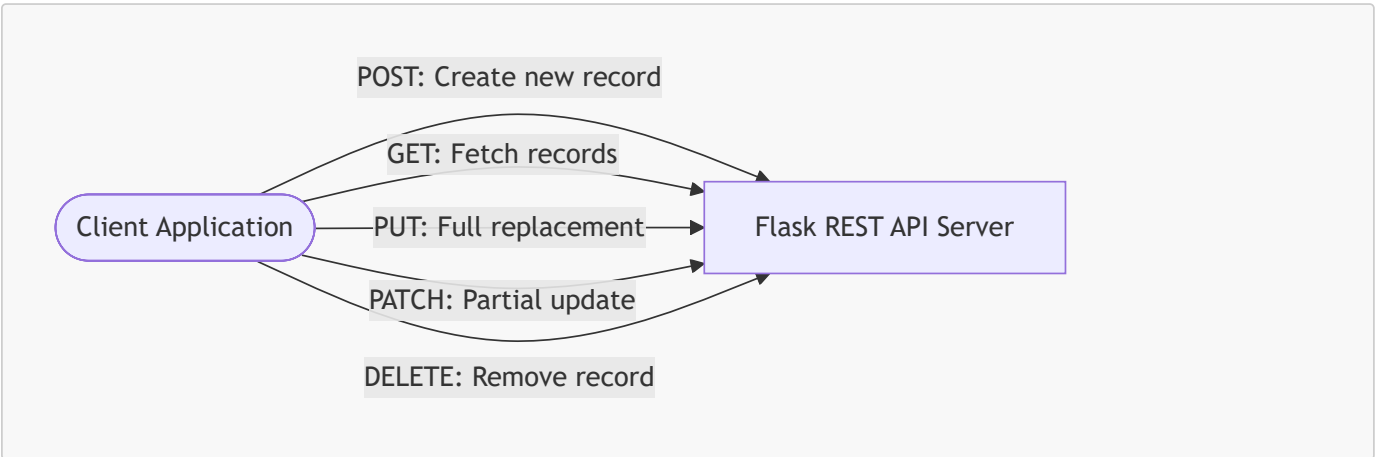
In REST, **URIs identify resources, not actions**. Resources must be modeled as **nouns**, never as verbs.

Good RESTful Design (Nouns, Pluralized)	Bad Design (RPC Style / Verbs in URL)	Explanation
<code>GET /api/v1/products</code>	<code>GET /api/v1/getAllProducts</code>	URIs identify <i>what</i> the resource is. HTTP verbs indicate <i>what to do</i> .
<code>GET /api/v1/products/42</code>	<code>GET /api/v1/getProductById?id=42</code>	Use path parameters for resource identity.
<code>POST /api/v1/products</code>	<code>POST /api/v1/createNewProduct</code>	POST implies resource creation. Don't repeat "create" in the path.
<code>PUT /api/v1/products/42</code>	<code>POST /api/v1/updateProduct/42</code>	Use HTTP PUT or PATCH to update.
<code>DELETE /api/v1/products/42</code>	<code>GET /api/v1/deleteProduct?id=42</code>	GET must be safe and read-only. Never mutate or delete via GET .

Good RESTful Design (Nouns, Pluralized)	Bad Design (RPC Style / Verbs in URL)	Explanation
GET /api/v1/orders/7/items	GET /api/v1/getOrderItems? order_id=7	Express hierarchical relationships naturally using nested paths.
[!TIP] API Versioning: Always prefix API routes with a version number (e.g., /api/v1/...). This allows you to publish breaking changes later under /api/v2/ without disrupting legacy client applications.		

2. HTTP Verbs in REST Semantics

REST maps CRUD (Create, Read, Update, Delete) operations directly to standard HTTP verbs:



- **GET:** Retrieve a resource or collection of resources. Query parameters (`?category=electronics&limit=10`) are used for filtering, pagination, and sorting.
- **POST:** Create a new subordinate resource. The request body contains the representation of the new resource. The server generates an ID and returns HTTP `201 Created`.
- **PUT:** Replace an existing resource in its entirety. The payload must provide the complete set of resource fields. If fields are omitted, the server assumes they should be wiped or set to defaults.
- **PATCH:** Apply a partial modification to a resource. Only the specific fields being changed need to be supplied in the request body.
- **DELETE:** Permanently remove the specified resource.

3. Safety and Idempotency Matrix

Two foundational concepts govern HTTP methods in REST:

- **Safe:** The method is strictly read-only and does not mutate the server state. Calling it causes no side-effects.
- **Idempotent:** Making N identical requests ($N \geq 1$) results in the exact same server state as making a single request.

HTTP Method	Safe?	Idempotent?	Request Body?	Standard Success Status
GET	Yes	Yes	No	200 OK
HEAD	Yes	Yes	No	200 OK (Headers only)
POST	No	No	Yes	201 Created
PUT	No	Yes	Yes	200 OK or 204 No Content
PATCH	No	No	Yes	200 OK or 204 No Content
DELETE	No	Yes	Optional	200 OK or 204 No Content

[!NOTE] **Why is DELETE idempotent?** The first `DELETE /api/products/10` removes product 10 (status `200` or `204`). A second `DELETE /api/products/10` may return `404 Not Found`, but the **state of the database** on the server is unchanged

—product 10 remains deleted. Hence, **DELETE** is idempotent.

4. Standard HTTP Status Codes in REST APIs

Status codes communicate the outcome of the request unambiguously to the client machine:

+-----+-----+-----+-----+			
1xx Informational	2xx Success	3xx Redirection	
4xx Client Error	5xx Server Error		
+-----+-----+-----+-----+			

1. 2xx Success:

- **200 OK**: Standard successful response for **GET**, **PUT**, **PATCH**, or **DELETE**.
- **201 Created**: Successfully created a new resource (via **POST**). The response should include the created object in the body and a **Location**: `/api/v1/products/42` header.
- **204 No Content**: Action succeeded, but the response body is intentionally empty (common for **DELETE** or **PUT**).

2. 4xx Client Errors (The client sent something incorrect):

- **400 Bad Request**: Malformed JSON syntax, invalid payload schema, or failing basic input validation.
- **401 Unauthorized**: Authentication is missing or invalid (e.g., missing API token).
- **403 Forbidden**: Authentication succeeded, but the client does not have permission to access or modify this specific resource.
- **404 Not Found**: The requested URI resource does not exist.
- **405 Method Not Allowed**: The endpoint exists, but the HTTP verb is unsupported (e.g., sending **POST** to a read-only endpoint).
- **409 Conflict**: Request cannot be fulfilled due to a business state conflict (e.g., creating a user with an email that is already registered).
- **422 Unprocessable Entity**: JSON syntax is valid, but internal validation failed (e.g., price is a negative number).

3. 5xx Server Errors (The server crashed or encountered an unhandled exception):

- **500 Internal Server Error**: Unhandled exception in Python code (database crashed, unhandled division by zero, null pointer).

Part 3: Core Flask Tools for REST APIs

1. Returning JSON: `jsonify` vs. `json.dumps`

In standard Python, `json.dumps(obj)` serializes a Python dictionary into a JSON string. However, in Flask web services, you should always use `flask.jsonify()`.

```
import json
from flask import Flask, Response, jsonify

app = Flask(__name__)

# Approach A: Raw json.dumps (Avoid in REST APIs)
@app.route("/raw-json")
def raw_json():
    data = {"status": "active", "code": 200}
    # Problem: Defaults to Content-Type: text/html!
    return json.dumps(data)

# Approach B: Flask jsonify (Standard REST approach)
@app.route("/api/status")
def api_status():
    data = {"status": "active", "code": 200}
```

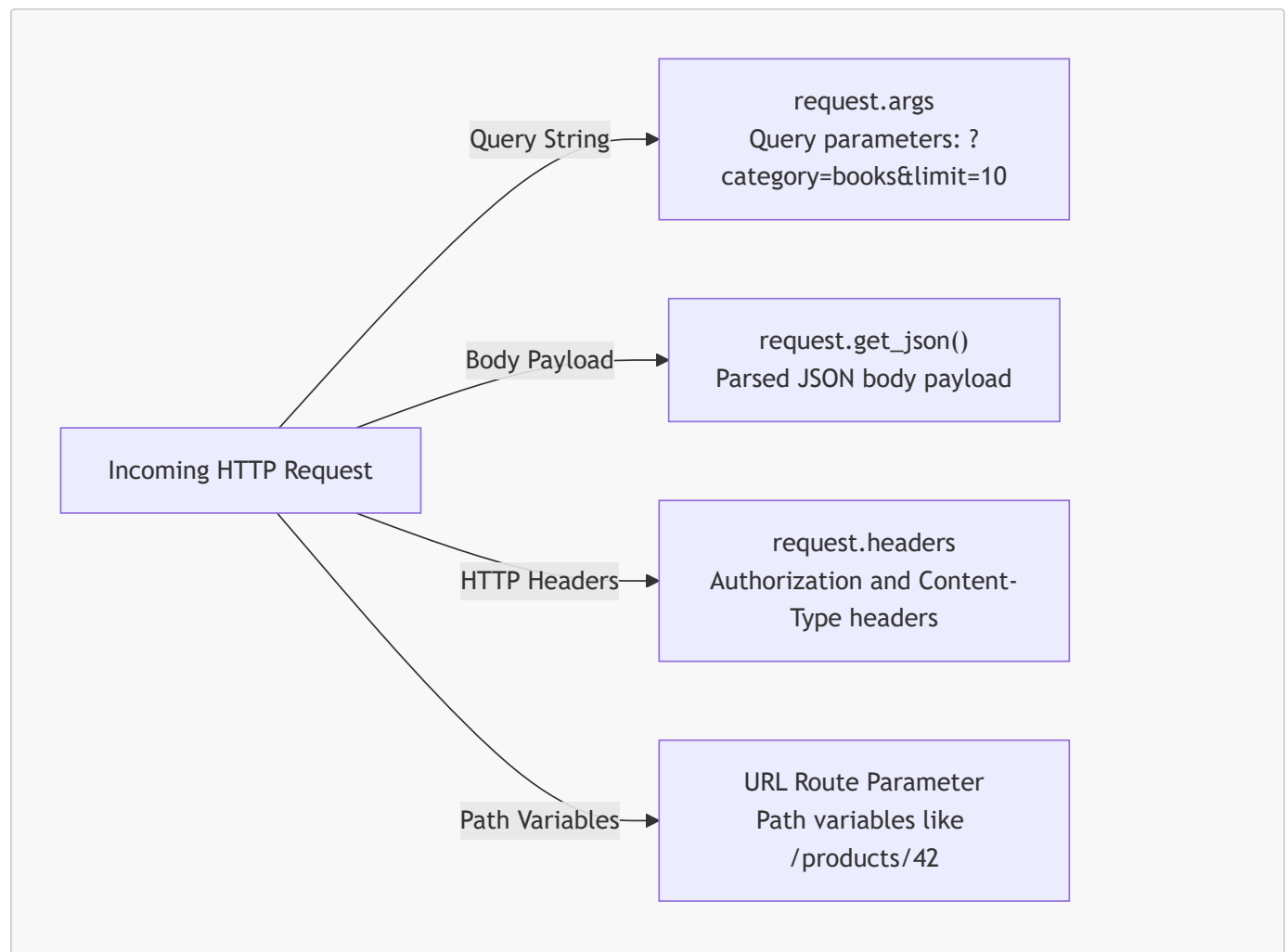
```
# Automatically sets Content-Type: application/json
return jsonify(data), 200
```

Why `jsonify` is Superior:

1. **MIME Type Header:** `jsonify()` automatically sets the HTTP response header `Content-Type: application/json`.
2. **Status Code and Header Tuples:** Flask allows returning a tuple `(jsonify(data), status_code, headers)`, making status code injection seamless.
3. **Configuration Aware:** `jsonify()` respects Flask configuration variables such as `JSON_SORT_KEYS` and custom JSON encoders.

2. Parsing Incoming Requests: `request.get_json()` and `request.args`

Flask provides the `request` context object to inspect different parts of the incoming HTTP transmission:



```
from flask import Flask, request, jsonify

app = Flask(__name__)

@app.route("/api/v1/products", methods=["GET", "POST"])
def manage_products():
    if request.method == "GET":
        # 1. Reading Query Parameters: ?category=electronics&limit=10
        category = request.args.get("category", default=None, type=str)
        limit = request.args.get("limit", default=20, type=int)

    return jsonify({
```

```
        "action": "list_products",
        "filter_category": category,
        "page_limit": limit
    )), 200

elif request.method == "POST":
    # 2. Reading JSON Body Payload
    # silent=True returns None instead of raising 400 Bad Request if JSON is malformed
    payload = request.get_json(silent=True)

    if payload is None:
        return jsonify({
            "error": "Bad Request",
            "message": "Request body must be valid application/json"
        }), 400

    # Validate required fields
    if "name" not in payload or "price" not in payload:
        return jsonify({
            "error": "Unprocessable Entity",
            "message": "Missing required fields: 'name' and 'price'"
        }), 422

    return jsonify({
        "message": "Product created successfully",
        "received_data": payload
    }), 201
```

[!IMPORTANT] Always use `request.get_json(silent=True)` or wrap `request.get_json()` inside a `try...except` block when accepting client input. By default, calling `request.get_json()` on a request that has an invalid JSON syntax or missing `Content-Type: application/json` header causes Flask to immediately trigger a raw HTTP 400 response.

3. Dynamic URL Parameters and Variable Converters

Flask allows capturing parts of the URL path directly into your view function using typed converter syntax `<converter:variable_name>`:

Converter	Matches	Example Route	Match	Reject
string (default)	Any text without slashes	@app.route("/users/<username>")	/users/john	/users/john/profile
int	Positive integers	@app.route("/api/products/<int:id>")	/api/products/42	/api/products/laptop
float	Positive floating point numbers	@app.route("/rates/<float:rate>")	/rates/3.14	/rates/abc
path	Accepts slashes	@app.route("/files/<path:filepath>")	/files/docs/readme.txt	(empty string)
uuid	UUID strings	@app.route("/orders/<uuid:order_id>")	/orders/123e4567-e89b...	/orders/99

```
@app.route("/api/v1/products/<int:product_id>", methods=["GET"])
def get_single_product(product_id: int):
```

```
# product_id is guaranteed to be a Python int
return jsonify({"product_id": product_id, "name": "Mechanical Keyboard"})
```

4. Centralized Error Handling with `@app.errorhandler`

In a REST API, responses must **never** return raw HTML error tracebacks or standard Apache/Nginx error templates. All errors—including 404 and 500—must be returned in a consistent, machine-readable JSON structure.

```
from flask import Flask, jsonify

app = Flask(__name__)

@app.errorhandler(404)
def not_found_handler(error):
    return jsonify({
        "success": False,
        "error": "Not Found",
        "message": "The requested resource endpoint does not exist."
    }), 404

@app.errorhandler(405)
def method_not_allowed_handler(error):
    return jsonify({
        "success": False,
        "error": "Method Not Allowed",
        "message": "The HTTP verb used is not permitted for this endpoint."
    }), 405

@app.errorhandler(500)
def internal_server_error_handler(error):
    return jsonify({
        "success": False,
        "error": "Internal Server Error",
        "message": "An unexpected error occurred on the server."
    }), 500
```

Part 4: Practical Project: Building a Products CRUD REST API with Flask & SQLite

Let us implement a complete, robust, production-grade RESTful API for an **Inventory Product Catalog** using Flask and SQLite.

1. Project Structure

```
inventory_api/
├── database.py    # SQLite connection and schema migration
├── app.py         # Flask application and REST endpoints
└── inventory.db  # SQLite database file (created automatically)
```

2. Database Schema & Helper Module (`database.py`)

Create `database.py`. This module handles database connections, sets up `sqlite3.Row` for dictionary-like column access, and initializes the `products` table.

```

"""
database.py - Database connection management and initialization.
"""

import sqlite3
from typing import Optional

DATABASE_NAME = "inventory.db"

def get_db_connection() -> sqlite3.Connection:
    """
    Creates and returns a thread-safe connection to the SQLite database.
    Configures row_factory to sqlite3.Row for key-based column lookup.
    """
    conn = sqlite3.connect(DATABASE_NAME)
    conn.row_factory = sqlite3.Row
    # Enable Foreign Key enforcement in SQLite
    conn.execute("PRAGMA foreign_keys = ON;")
    return conn

def init_db() -> None:
    """
    Initializes the database schema if tables do not already exist.
    """
    schema_sql = """
    CREATE TABLE IF NOT EXISTS products (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        name TEXT NOT NULL,
        category TEXT NOT NULL,
        price REAL NOT NULL CHECK (price >= 0),
        stock INTEGER NOT NULL DEFAULT 0 CHECK (stock >= 0),
        created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    );
    """
    with get_db_connection() as conn:
        conn.executescript(schema_sql)
        conn.commit()

if __name__ == "__main__":
    init_db()
    print("Database schema initialized successfully.")

```

3. The REST API Application (app.py)

Create `app.py`. This implements full CRUD capabilities:

- GET `/api/v1/products` - List all products with optional query filtering (`?category=...&min_price=...`)
- GET `/api/v1/products/<int:id>` - Retrieve a single product by ID
- POST `/api/v1/products` - Create a new product with complete validation
- PUT `/api/v1/products/<int:id>` - Full replacement update of a product
- PATCH `/api/v1/products/<int:id>` - Partial field update of a product
- DELETE `/api/v1/products/<int:id>` - Delete a product

```

"""
app.py - Production-ready Flask RESTful API for Product Inventory.

```

```

"""

import sqlite3
from flask import Flask, jsonify, request
from database import get_db_connection, init_db

app = Flask(__name__)

# Initialize database schema upon application launch
init_db()

# -----
# Helper Functions
# -----

def row_to_dict(row: sqlite3.Row) -> dict:
    """Converts an sqlite3.Row object into a serializable standard Python dictionary."""
    return dict(row)

def json_response(data=None, message: str = "", status_code: int = 200, success: bool = True):
    """Utility helper to produce uniform, standardized API response envelopes."""
    payload = {
        "success": success,
        "status_code": status_code
    }
    if message:
        payload["message"] = message
    if data is not None:
        payload["data"] = data
    return jsonify(payload), status_code

# -----
# Centralized Error Handlers
# -----

@app.errorhandler(404)
def handle_404(error):
    return json_response(
        message="The requested endpoint or resource was not found.",
        status_code=404,
        success=False
    )

@app.errorhandler(405)
def handle_405(error):
    return json_response(
        message=f"HTTP verb '{request.method}' is not allowed for this route.",
        status_code=405,
        success=False
    )

@app.errorhandler(500)
def handle_500(error):
    return json_response(
        message="An unexpected server error occurred.",
        status_code=500,
        success=False
    )

```

```

# -----
# REST API Endpoints: /api/v1/products
# -----

@app.route("/api/v1/products", methods=["GET"])
def get_products():
    """
    GET /api/v1/products
    Retrieves all products.
    Supports query parameters for filtering:
        - ?category=<name>
        - ?min_price=<value>
        - ?limit=<number>
    """
    category = request.args.get("category", default=None, type=str)
    min_price = request.args.get("min_price", default=None, type=float)
    limit = request.args.get("limit", default=50, type=int)

    query = "SELECT * FROM products WHERE 1=1"
    params = []

    if category:
        query += " AND LOWER(category) = LOWER(?)"
        params.append(category)

    if min_price is not None:
        query += " AND price >= ?"
        params.append(min_price)

    query += " ORDER BY id DESC LIMIT ?"
    params.append(limit)

    with get_db_connection() as conn:
        cursor = conn.cursor()
        cursor.execute(query, tuple(params))
        rows = cursor.fetchall()

    products = [row_to_dict(row) for row in rows]
    return json_response(data=products, status_code=200)

@app.route("/api/v1/products/<int:product_id>", methods=["GET"])
def get_product(product_id: int):
    """
    GET /api/v1/products/<id>
    Retrieves a single product by its primary key ID.
    Returns 404 if not found.
    """
    with get_db_connection() as conn:
        cursor = conn.cursor()
        cursor.execute("SELECT * FROM products WHERE id = ?", (product_id,))
        row = cursor.fetchone()

    if row is None:
        return json_response(
            message=f"Product with ID {product_id} does not exist.",
            status_code=404,
            success=False
        )

    return json_response(data=row_to_dict(row), status_code=200)

```

```

@app.route("/api/v1/products", methods=["POST"])
def create_product():
    """
    POST /api/v1/products
    Creates a new product record.
    Required JSON fields: 'name', 'category', 'price', 'stock'
    """
    body = request.get_json(silent=True)
    if body is None:
        return json_response(
            message="Request body must be a valid JSON object with 'Content-Type:
application/json'.",
            status_code=400,
            success=False
        )

    # Validate presence of required attributes
    required_fields = ["name", "category", "price", "stock"]
    missing_fields = [f for f in required_fields if f not in body]
    if missing_fields:
        return json_response(
            message=f"Missing required fields: {' '.join(missing_fields)}",
            status_code=422,
            success=False
        )

    name = str(body["name"]).strip()
    category = str(body["category"]).strip()

    try:
        price = float(body["price"])
        stock = int(body["stock"])
        if price < 0 or stock < 0:
            raise ValueError()
    except (ValueError, TypeError):
        return json_response(
            message="'price' must be a non-negative float and 'stock' must be a non-negative
integer.",
            status_code=422,
            success=False
        )

    if not name or not category:
        return json_response(
            message="'name' and 'category' cannot be empty strings.",
            status_code=422,
            success=False
        )

    with get_db_connection() as conn:
        cursor = conn.cursor()
        cursor.execute(
            "INSERT INTO products (name, category, price, stock) VALUES (?, ?, ?, ?)",
            (name, category, price, stock)
        )
        conn.commit()
        new_id = cursor.lastrowid

        cursor.execute("SELECT * FROM products WHERE id = ?", (new_id,))
        new_product = row_to_dict(cursor.fetchone())

```

```

# Build response with 201 Created and Location header
response = jsonify({
    "success": True,
    "status_code": 201,
    "message": "Product created successfully.",
    "data": new_product
})
response.status_code = 201
response.headers["Location"] = f"/api/v1/products/{new_id}"
return response

@app.route("/api/v1/products/<int:product_id>", methods=["PUT"])
def replace_product(product_id: int):
    """
    PUT /api/v1/products/<id>
    Idempotent complete replacement of a product record.
    All required fields must be supplied.
    """
    body = request.get_json(silent=True)
    if body is None:
        return json_response(
            message="Invalid or missing JSON payload.",
            status_code=400,
            success=False
        )

    # Full replacement requires all mandatory fields
    required_fields = ["name", "category", "price", "stock"]
    missing = [f for f in required_fields if f not in body]
    if missing:
        return json_response(
            message=f"PUT requires all fields for complete resource replacement: {'',
            '.join(missing)}",
            status_code=422,
            success=False
        )

    try:
        price = float(body["price"])
        stock = int(body["stock"])
        if price < 0 or stock < 0:
            raise ValueError()
    except (ValueError, TypeError):
        return json_response(
            message="Invalid numeric values for price or stock.",
            status_code=422,
            success=False
        )

    with get_db_connection() as conn:
        cursor = conn.cursor()
        cursor.execute("SELECT id FROM products WHERE id = ?", (product_id,))
        if cursor.fetchone() is None:
            return json_response(
                message=f"Product with ID {product_id} not found.",
                status_code=404,
                success=False
            )

        cursor.execute(
            """
            UPDATE products

```

```

        SET name = ?, category = ?, price = ?, stock = ?
        WHERE id = ?
        """
        (body["name"], body["category"], price, stock, product_id)
    )
    conn.commit()

    cursor.execute("SELECT * FROM products WHERE id = ?", (product_id,))
    updated = row_to_dict(cursor.fetchone())

    return json_response(data=updated, message="Product replaced successfully.", status_code=200)

@app.route("/api/v1/products/<int:product_id>", methods=["PATCH"])
def patch_product(product_id: int):
    """
    PATCH /api/v1/products/<id>
    Partial update. Only the fields present in the request body are updated.
    """
    body = request.get_json(silent=True)
    if not body or not isinstance(body, dict):
        return json_response(
            message="Valid JSON payload required for partial update.",
            status_code=400,
            success=False
        )

    allowed_fields = {"name", "category", "price", "stock"}
    update_fields = [k for k in body.keys() if k in allowed_fields]

    if not update_fields:
        return json_response(
            message=f"No valid update fields supplied. Allowed: {allowed_fields}",
            status_code=422,
            success=False
        )

    # Build dynamic SQL update string
    set_clauses = []
    params = []
    for field in update_fields:
        val = body[field]
        if field == "price":
            try:
                val = float(val)
                if val < 0:
                    raise ValueError()
            except (ValueError, TypeError):
                return json_response(message="Price must be a non-negative float.",
status_code=422, success=False)
            elif field == "stock":
                try:
                    val = int(val)
                    if val < 0:
                        raise ValueError()
                except (ValueError, TypeError):
                    return json_response(message="Stock must be a non-negative integer.",
status_code=422, success=False)

            set_clauses.append(f"{field} = ?")
            params.append(val)

    params.append(product_id)

```

```

sql = f"UPDATE products SET {' '.join(set_clauses)} WHERE id = ?"

with get_db_connection() as conn:
    cursor = conn.cursor()
    cursor.execute("SELECT id FROM products WHERE id = ?", (product_id,))
    if cursor.fetchone() is None:
        return json_response(
            message=f"Product with ID {product_id} not found.",
            status_code=404,
            success=False
        )

    cursor.execute(sql, tuple(params))
    conn.commit()

    cursor.execute("SELECT * FROM products WHERE id = ?", (product_id,))
    updated = row_to_dict(cursor.fetchone())

    return json_response(data=updated, message="Product patched successfully.", status_code=200)

@app.route("/api/v1/products/<int:product_id>", methods=["DELETE"])
def delete_product(product_id: int):
    """
    DELETE /api/v1/products/<id>
    Idempotently deletes a product record.
    """
    with get_db_connection() as conn:
        cursor = conn.cursor()
        cursor.execute("SELECT id FROM products WHERE id = ?", (product_id,))
        row = cursor.fetchone()

        if row is None:
            return json_response(
                message=f"Product with ID {product_id} not found.",
                status_code=404,
                success=False
            )

        cursor.execute("DELETE FROM products WHERE id = ?", (product_id,))
        conn.commit()

        return json_response(message=f"Product with ID {product_id} has been deleted.",
            status_code=200)

if __name__ == "__main__":
    # In development: debug=True allows hot-reload and informative terminal traces
    app.run(host="127.0.0.1", port=5000, debug=True)

```

4. Testing Endpoints with `curl`

To test your REST API without a frontend, use `curl` from your terminal:

1. Create a Product (POST):

```

curl -X POST http://127.0.0.1:5000/api/v1/products \
  -H "Content-Type: application/json" \
  -d '{
    "name": "Wireless Mechanical Keyboard",

```

```
    "category": "Electronics",
    "price": 89.99,
    "stock": 35
  }
```

Response (HTTP 201 Created):

```
{
  "data": {
    "category": "Electronics",
    "created_at": "2026-09-07 10:15:32",
    "id": 1,
    "name": "Wireless Mechanical Keyboard",
    "price": 89.99,
    "stock": 35
  },
  "message": "Product created successfully.",
  "status_code": 201,
  "success": true
}
```

2. Retrieve All Products with Filter (GET):

```
curl -X GET "http://127.0.0.1:5000/api/v1/products?category=Electronics&min_price=50"
```

3. Retrieve Single Product by ID (GET):

```
curl -X GET http://127.0.0.1:5000/api/v1/products/1
```

4. Partial Update of Stock (PATCH):

```
curl -X PATCH http://127.0.0.1:5000/api/v1/products/1 \
  -H "Content-Type: application/json" \
  -d '{"stock": 40}'
```

5. Delete Product (DELETE):

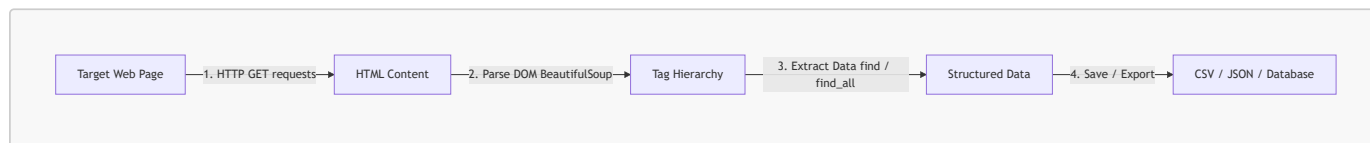
```
curl -X DELETE http://127.0.0.1:5000/api/v1/products/1
```

SECTION B: Web Scraping Basics in Python

1. What is Web Scraping?

Web scraping is the automated process of extracting data from website HTML pages. While REST APIs (covered in Section A) deliver structured JSON data cleanly, many websites do not provide a public API. Web scraping bridges this gap by:

1. Sending an HTTP request to download the target web page.
2. Parsing the resulting HTML document into a searchable tree structure (DOM).
3. Searching and extracting targeted elements (headings, article text, table data, links).
4. Saving or processing the data into structured formats (lists, dictionaries, CSV files).



2. Key Libraries: `requests` and `BeautifulSoup`

Python provides two industry-standard libraries for web scraping:

```
pip install requests beautifulsoup4
```

- **`requests`**: Handles HTTP networking. It sends requests (`GET`, `POST`) and fetches the server's HTML response.
- **`beautifulsoup4` (`bs4`)**: A powerful HTML parser that navigates and queries elements using tag names, CSS classes, IDs, and attributes.

3. Core Scraping Workflow & Methods

A. Fetching and Creating the "Soup"

```
import requests
from bs4 import BeautifulSoup

url = "http://quotes.toscrape.com/"
headers = {"User-Agent": "Mozilla/5.0 (Educational Scraping Demo)"}

# 1. Fetch web page
response = requests.get(url, headers=headers, timeout=10)
response.raise_for_status() # Raises an exception if the HTTP request failed (e.g. 404, 500)

# 2. Parse HTML into a BeautifulSoup object
soup = BeautifulSoup(response.text, "html.parser")
```

B. Searching Elements

Method	Purpose	Example
<code>soup.find("tag")</code>	Finds the first matching element	<code>first_h1 = soup.find("h1")</code>
<code>soup.find_all("tag")</code>	Finds all matching elements (returns list)	<code>all_paragraphs = soup.find_all("p")</code>
<code>soup.find("tag", class_="name")</code>	Finds by CSS class (use <code>class_</code>)	<code>card = soup.find("div", class_="quote")</code>
<code>soup.find("tag", id="name")</code>	Finds by element ID	<code>header = soup.find("nav", id="navbar")</code>
<code>soup.select("css selector")</code>	Finds using standard CSS selectors	<code>items = soup.select(".list-group > li.item")</code>

C. Extracting Text and Attributes

```
# Extract clean inner text (.strip removes surrounding whitespace)
heading_text = soup.find("h1").get_text(strip=True)

# Extract HTML attributes (href, src, alt, etc.)
first_link = soup.find("a")
href_url = first_link.get("href")
```

4. Practical Scraping Example

The following script demonstrates scraping quotes, authors, and topic tags from **Quotes to Scrape** (a public sandbox site built specifically for scraping practice):

```
"""
web_scraping_demo.py - Scrape quotes, authors, and tags using requests and BeautifulSoup.
"""

import requests
from bs4 import BeautifulSoup

def scrape_quotes():
    url = "http://quotes.toscrape.com/"
    headers = {"User-Agent": "PythonStudentScraper/1.0"}

    print(f"Fetching {url} ...")
    response = requests.get(url, headers=headers, timeout=10)
    response.raise_for_status()

    # Parse HTML
    soup = BeautifulSoup(response.text, "html.parser")

    # Find all quote container blocks (<div class="quote">)
    quote_blocks = soup.find_all("div", class_="quote")
    print(f"Found {len(quote_blocks)} quotes on the page.\n")

    results = []
    for block in quote_blocks:
        # Extract quote text (<span class="text">)
        text = block.find("span", class_="text").get_text(strip=True)

        # Extract author name (<small class="author">)
        author = block.find("small", class_="author").get_text(strip=True)

        # Extract tags (<a class="tag">)
        tag_elements = block.find_all("a", class_="tag")
        tags = [t.get_text(strip=True) for t in tag_elements]

        results.append({
            "quote": text,
            "author": author,
            "tags": tags
        })

    return results

if __name__ == "__main__":
    quotes = scrape_quotes()
```

```
# Display the first 3 scraped quotes
for i, item in enumerate(quotes[:3], 1):
    print(f"{i}. \"{item['quote']}\"")
    print(f"    - Author: {item['author']}")
    print(f"    - Tags : {' '.join(item['tags'])}\n")
```

Execution Output:

```
Fetching http://quotes.toscrape.com/ ...
Found 10 quotes on the page.
```

1. "“The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking.”"
- Author: Albert Einstein
- Tags : change, deep-thoughts, thinking, world
2. "“It is our choices, Harry, that show what we truly are, far more than our abilities.”"
- Author: J.K. Rowling
- Tags : abilities, choices
3. "“There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle.”"
- Author: Albert Einstein
- Tags : inspirational, life, live, miracle, miracles

5. Best Practices & Ethical Scraping

1. **Inspect robots.txt:** Check <https://website.com/robots.txt> before scraping to see which paths are disallowed for automated bots.
2. **Set a Custom User-Agent:** Identify your script in request headers so server administrators know who is requesting data.
3. **Rate Limiting:** Use `time.sleep(1)` between multiple page requests to avoid overwhelming target servers.
4. **Prefer Official APIs:** If a website provides a free or official REST API, always use the API instead of web scraping.
5. **Handle Network Exceptions:** Wrap requests in `try / except requests.RequestException` and always include a `timeout` argument.

SECTION C: Introduction to NumPy & Image Manipulation

1. Introduction to NumPy: Capabilities, Applications & List Comparison

A. What is NumPy?

NumPy (short for **Numerical Python**) is the core library for scientific, numerical, and data-driven computing in Python. Created in 2005 by Travis Oliphant, it provides the foundational data structure that underpins virtually the entire modern Python data science and machine learning ecosystem: the `ndarray` (N -dimensional array).

While Python is an expressive, dynamic language prized for developer productivity, pure Python loops are notoriously slow when processing millions of arithmetic operations. NumPy overcomes this bottleneck by executing numerical operations using pre-compiled, optimized **C and Fortran routines** under the hood, while exposing an intuitive, high-level Python API.

```
pip install numpy pillow
```

B. Core Capabilities of NumPy

- 1. ***N*-Dimensional Array (ndarray)**: Fast, memory-efficient multi-dimensional arrays supporting 1D vectors, 2D matrices, 3D image arrays, and higher-dimensional tensors.
- 2. **Vectorization**: Perform math operations on entire arrays simultaneously without writing explicit Python `for` loops.
- 3. **Broadcasting**: Automatically compute arithmetic between arrays of different but compatible dimensions (e.g. adding a 1D vector across all rows of a 2D matrix, or applying a 3-channel color multiplier to an entire image).
- 4. **Universal Functions (ufuncs)**: Fast element-wise mathematical functions (e.g., `np.sin`, `np.exp`, `np.log`, `np.sqrt`, `np.clip`) executed directly in compiled C.
- 5. **Linear Algebra**: Native support for matrix multiplication (`@`), dot products, inversions, eigenvalues, and decompositions.
- 6. **Random Number Generation**: Fast generation of statistical distributions (normal, uniform, binomial) using modern pseudo-random BitGenerators.

C. Real-World Fields of Application

NumPy is the foundational bedrock upon which the modern Python data ecosystem is built:

Field	How NumPy is Applied	Key Libraries Powered by NumPy
Data Science & Analytics	Tabular data manipulation, statistical aggregations, missing-data handling, time-series analysis.	Pandas, Polars, Statsmodels
Machine Learning & AI	Feature vectors, gradient computations, model weight matrices. PyTorch and TensorFlow tensors are modeled directly after NumPy's <code>ndarray</code> .	scikit-learn, PyTorch, TensorFlow
Computer Vision & Image Processing	Digital images and video frames are stored and processed directly as 2D/3D pixel matrices.	OpenCV, Pillow (PIL), scikit-image
Audio & Signal Processing	Digital audio waveforms, sampling rates, frequencies, and Fast Fourier Transforms (FFT).	SciPy (signal), Librosa
Quantitative Finance	Algorithmic trading, portfolio optimization, options pricing (Monte Carlo / Black-Scholes models), risk management.	QuantLib, TA-Lib
Physics & Engineering Simulations	Fluid dynamics, weather forecasting, molecular dynamics, and astronomical imaging (e.g., the Event Horizon Telescope black hole photo).	SciPy, Astropy, Biopython

D. NumPy ndarray vs. Python List: Why Not Just Use Lists?

At first glance, a Python `list` might seem similar to a NumPy array. However, their internal memory architecture and performance characteristics are fundamentally different:

Python List Memory Layout (Array of Pointers):

List Object -> [Ptr 1 | Ptr 2 | Ptr 3 | Ptr 4]

|

|

|

|

v

v

v

v

[PyObject] [PyObject] [PyObject] [PyObject] (Scattered across Heap memory!)

NumPy ndarray Memory Layout (Contiguous C Buffer):

ndarray Object -> [Value 1 | Value 2 | Value 3 | Value 4] (Packed tightly in one contiguous block!)

Feature	Python Standard <code>list</code>	NumPy <code>ndarray</code>
Memory Layout	Scattered: An array of pointers referencing independent Python objects scattered across heap memory.	Contiguous: Elements are stored back-to-back in a single, unbroken block of memory (C-style layout).
Data Types	Heterogeneous: Can hold integers, strings, floats, and objects within the same list.	Homogeneous: Every element has the exact same data type (e.g. all <code>int32</code> , <code>float64</code> , or <code>uint8</code>).
Memory Overhead	High: Each number is a full Python object (~28 bytes for an integer + 8 bytes pointer = ~36 bytes per number).	Minimal: Stored as raw binary bytes (e.g., <code>uint8</code> = exactly 1 byte per value).
Execution Speed	Slow: Loops must dereference pointers and perform dynamic type checking on every iteration.	Blazing Fast: Leverages CPU cache locality, SIMD (Single Instruction Multiple Data), and C-level execution.
Math Operations	<code>list + [5]</code> appends element <code>5</code> to the list; <code>list * 2</code> duplicates the list contents.	<code>arr + 5</code> adds 5 to every element ; <code>arr * 2</code> doubles every value .
Multi-Dimensional Indexing	Nested indexing: <code>matrix[row][col]</code>	Clean matrix coordinate indexing: <code>arr[row, col]</code> or <code>img[y, x, channel]</code>

E. Performance Benchmark: List vs. NumPy

Run this short benchmark to observe the real-world performance difference between a Python list loop and NumPy vectorization on 1,000,000 numbers:

```
"""
benchmark_list_vs_numpy.py - Comparing execution speed of Python list vs. NumPy array.
"""

import time
import numpy as np

SIZE = 1_000_000

# 1. Python List: Element-wise doubling via list comprehension
py_list = list(range(SIZE))
start = time.time()
list_result = [x * 2 for x in py_list]
list_time = time.time() - start

# 2. NumPy Array: Element-wise doubling via vectorization
np_arr = np.arange(SIZE)
start = time.time()
np_result = np_arr * 2
numpy_time = time.time() - start

print(f"Python List time : {list_time:.4f} seconds")
print(f"NumPy Array time : {numpy_time:.4f} seconds")
print(f"-> NumPy is {list_time / numpy_time:.1f}x faster!")
```

Typical Output:

```
Python List time : 0.0465 seconds
NumPy Array time : 0.0036 seconds
-> NumPy is 12.7x faster!
```

2. Digital Images as NumPy Arrays

Every digital picture is simply a multi-dimensional grid of numbers:

- **Grayscale Image (2D Matrix):**
 - Shape: (Height, Width)
 - Each element is an integer from 0 (black) to 255 (white).
- **Color Image (3D Array - RGB):**
 - Shape: (Height, Width, 3)
 - Channel 0: **Red** (0 to 255)
 - Channel 1: **Green** (0 to 255)
 - Channel 2: **Blue** (0 to 255)
- **Data Type:** Digital images use `np.uint8` (unsigned 8-bit integers, $0 \leq \text{value} \leq 255$).

```
Image Coordinate System:
(0,0) -----> Column X (Width, Axis 1)
|
|      Pixel [y, x] = [Red, Green, Blue] (Axis 2)
|      v
v      Row Y (Height, Axis 0)
```

The Python imaging library **Pillow (PIL)** bridges images on disk with NumPy arrays:

- **Disk** → **NumPy**: `img = np.array(Image.open("photo.jpg"))`
- **NumPy** → **Disk**: `Image.fromarray(img).save("output.jpg")`

3. Loading & Inspecting an Image

```
from PIL import Image
import numpy as np

# 1. Load an image file into a NumPy array
# (You can use any JPG or PNG image on your computer)
img = np.array(Image.open("sample.jpg"))

# 2. Inspect core array attributes
print("Data Type :", img.dtype) # uint8 (values from 0 to 255)
print("Dimensions:", img.ndim)  # 3 (Height, Width, Color Channels)
print("Shape      :", img.shape) # e.g., (300, 400, 3) -> 300 rows, 400 cols, 3 channels
print("Pixel(0,0):", img[0, 0]) # [R, G, B] values of the top-left pixel
```

4. Image Manipulation Examples

Because an image is just a NumPy array, you can manipulate it using basic array slicing, indexing, and vectorized math:

A. Cropping (2D Slicing)

Extract a sub-region using standard Python slice notation `[ymin:ymax, xmin:xmax]`:

```
# Crop a region: rows 50 to 200, columns 100 to 300
# Tip: Use .copy() to create an independent array
cropped = img[50:200, 100:300].copy()
Image.fromarray(cropped).save("cropped.jpg")
```

B. Flipping (Reversing Axes)

Mirror or turn an image upside-down by reversing array indices (`[::-1]`):

```
# Horizontal Flip (Mirror: reverse columns along Axis 1)
horizontal_flip = img[:, ::-1]
Image.fromarray(horizontal_flip).save("flipped_horizontal.jpg")

# Vertical Flip (Upside-down: reverse rows along Axis 0)
vertical_flip = img[::-1, :]
Image.fromarray(vertical_flip).save("flipped_vertical.jpg")
```

C. Adjusting Brightness & Tone (Vectorized Math & Clamping)

To increase brightness or adjust color channels, add or multiply values across the array.

[!WARNING] In `uint8` math, values wrap around if they exceed 255 ($240 + 30 = 270 \rightarrow 14$). Always clamp values between 0 and 255 using `np.clip()`:

```
# 1. Increase Brightness (+40):
brightened = np.clip(img.astype(np.int16) + 40, 0, 255).astype(np.uint8)
Image.fromarray(brightened).save("brightened.jpg")

# 2. Warm Sunset Tint (Boost Red, soften Blue via broadcasting):
warm_filter = np.array([1.25, 1.05, 0.75])
warm_img = np.clip(img * warm_filter, 0, 255).astype(np.uint8)
Image.fromarray(warm_img).save("warm_tint.jpg")
```

D. Converting to Grayscale

Use the standard human eye perceptual luminance formula ($0.299R + 0.587G + 0.114B$):

```
# Convert 3D RGB array to 2D Grayscale matrix
gray = (
    img[:, :, 0] * 0.299 +
    img[:, :, 1] * 0.587 +
    img[:, :, 2] * 0.114
).astype(np.uint8)

Image.fromarray(gray).save("grayscale.jpg")
print("Grayscale Shape:", gray.shape) # 2D array: (Height, Width)
```

5. Complete Image Processing Script

Here is a complete, self-contained Python script demonstrating loading, manipulating, and saving images. If you do not have an image ready, it automatically creates a starter `sample.jpg`:

```

"""
numpy_image_demo.py - Basic image loading and manipulation with NumPy and Pillow.
"""

import os
import numpy as np
from PIL import Image, ImageDraw

def get_or_create_sample_image(filename: str = "sample.jpg") -> None:
    """Creates a starter sample image if one does not already exist."""
    if not os.path.exists(filename):
        img = Image.new("RGB", (400, 300), color=(135, 206, 235)) # Sky blue
        draw = ImageDraw.Draw(img)
        draw.rectangle([0, 180, 400, 300], fill=(34, 139, 34)) # Green grass
        draw.ellipse([280, 40, 360, 120], fill=(255, 215, 0)) # Sun
        img.save(filename)
        print(f"Created starter '{filename}'")

def main():
    # Ensure a sample image exists to work with
    get_or_create_sample_image("sample.jpg")

    # 1. Load image from disk into a NumPy array
    img = np.array(Image.open("sample.jpg"))
    h, w, c = img.shape
    print(f"Loaded image: {w}x{h} pixels, shape: {img.shape}, dtype: {img.dtype}")

    # 2. Crop center region
    cropped = img[h // 4 : 3 * h // 4, w // 4 : 3 * w // 4].copy()
    Image.fromarray(cropped).save("output_cropped.jpg")

    # 3. Horizontal mirror flip
    flipped = img[:, ::-1]
    Image.fromarray(flipped).save("output_flipped.jpg")

    # 4. Warm sunset color tint (broadcasting)
    warm = np.clip(img * [1.25, 1.05, 0.75], 0, 255).astype(np.uint8)
    Image.fromarray(warm).save("output_warm.jpg")

    # 5. Grayscale conversion
    gray = (img[:, :, 0] * 0.299 + img[:, :, 1] * 0.587 + img[:, :, 2] * 0.114).astype(np.uint8)
    Image.fromarray(gray).save("output_gray.jpg")

    # 6. Side-by-side comparison (Horizontal stack)
    comparison = np.hstack((img, warm))
    Image.fromarray(comparison).save("output_comparison.jpg")

    print("\nSuccessfully processed and saved:")
    print(" - output_cropped.jpg")
    print(" - output_flipped.jpg")
    print(" - output_warm.jpg")
    print(" - output_gray.jpg")
    print(" - output_comparison.jpg")

if __name__ == "__main__":
    main()

```

Quick Reference: Common Image Operations

Operation	NumPy Code	Purpose
Load Image	<code>np.array(Image.open("pic.jpg"))</code>	Converts image to 3D <code>uint8</code> array.
Save Image	<code>Image.fromarray(arr).save("out.jpg")</code>	Converts <code>uint8</code> array back to image file.
Crop	<code>img[y1:y2, x1:x2].copy()</code>	Extracts bounding box region.
Horizontal Flip	<code>img[:, ::-1]</code>	Mirrors left-to-right.
Vertical Flip	<code>img[:, :, ::-1]</code>	Flips upside-down.
Brightness	<code>np.clip(img.astype(np.int16) + 40, 0, 255).astype(np.uint8)</code>	Safely brightens pixels without overflow.
Grayscale	<code>(R*0.299 + G*0.587 + B*0.114).astype(np.uint8)</code>	Converts 3D RGB to 2D perceptual grayscale.
Stack Side-by-Side	<code>np.hstack((img1, img2))</code>	Places two images together for comparison.